

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – [AI Agent for Digital Financial Literacy]

Domain of Project – [AI-powered financial literacy assistant that helps users understand digital payments, online banking, budgeting, loans, and fraud prevention through smart multilingual guidance.]

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Problem Statement -

Problem Statement : AI Agent for Digital Financial Literacy

The Challenge

Digital financial services such as UPI payments, online banking, digital wallets, loans, and investment platforms are rapidly growing, but many individuals still lack the financial knowledge and awareness needed to use them safely and effectively. Information related to digital payments, fraud prevention, budgeting, interest rates, savings, and personal finance is often scattered across unreliable or difficult-to-understand sources. Without personalized and real-time financial guidance, users may fall victim to scams, misuse financial tools, or struggle to make informed financial decisions.

The Objective

Build an AI-powered Digital Financial Literacy Agent that delivers personalized financial guidance and education for individuals, students, families, and small business users. The solution should:

Personalized Financial Guidance

Recommend suitable financial practices based on user income, spending habits, savings goals, financial literacy level, and banking preferences.

Financial Data RAG Retrieval

Fetch and summarize trusted information related to UPI, banking services, loans, insurance, budgeting, and digital finance from government portals, banking websites, and financial education platforms.

Expense Tracking & Budget Feedback

Allow users to log expenses via text, voice, or uploaded receipts and provide instant analysis of spending patterns, savings opportunities, and budgeting tips.

Fraud Prevention & Financial Safety

Provide scam alerts, fraud awareness tips, secure online transaction guidance, and safe digital banking practices to protect users from cyber fraud and financial scams.

Multi-Agent System

- Financial Knowledge Agent (retrieves trusted financial information)
- Budget Recommendation Agent (suggests personalized budgeting and savings plans)
- Fraud Detection & Safety Agent (provides scam prevention guidance)
- Expense Tracking & Feedback Agent (analyzes user spending and financial habits)

Visualization Dashboard

Display daily/monthly expenses, savings trends, spending categories, budgeting performance, loan calculations, and financial progress through interactive charts and dashboards.

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Technology Used

Technology Used:
IBM Cloud Lite Services + IBM Granite Models with Retrieval-Augmented Generation (RAG) for intelligent financial assistance, multilingual support, secure financial guidance, and real-time information retrieval.

Proposed Solution -

Proposed Solution: AI-Powered Digital Financial Literacy Agent

The proposed solution is a multi-agent AI-powered Digital Financial Literacy Agent built using LangFlow, IBM watsonx.ai, and Granite Models to provide personalized, real-time financial guidance and awareness. The system addresses challenges related to digital payments, online fraud, budgeting, banking literacy, and personal finance management through an intelligent and modular architecture.

At its core, the solution uses a Retrieval-Augmented Generation (RAG) pipeline to fetch accurate financial information from trusted sources such as government portals, banking websites, RBI guidelines, and financial education platforms. This information is embedded and stored in a vector database, enabling the system to retrieve and summarize financial concepts such as UPI payments, loan interest rates, budgeting strategies, fraud prevention tips, and digital banking practices using Granite models.

The system consists of four key agents:

Financial Knowledge Agent retrieves and summarizes trusted financial information using RAG.

Budget Recommendation Agent generates personalized budgeting and savings recommendations based on user income, expenses, and financial goals.

Fraud Detection & Safety Agent provides scam awareness tips, secure transaction guidance, and online fraud prevention recommendations.

Expense Tracking & Feedback Agent allows users to log expenses through text or voice and delivers instant spending analysis with actionable financial insights.

The solution includes a user-friendly dashboard (built with Streamlit or React) that visualizes daily and monthly expenses, savings patterns, budgeting performance, loan calculations, and financial progress through interactive charts and reports.

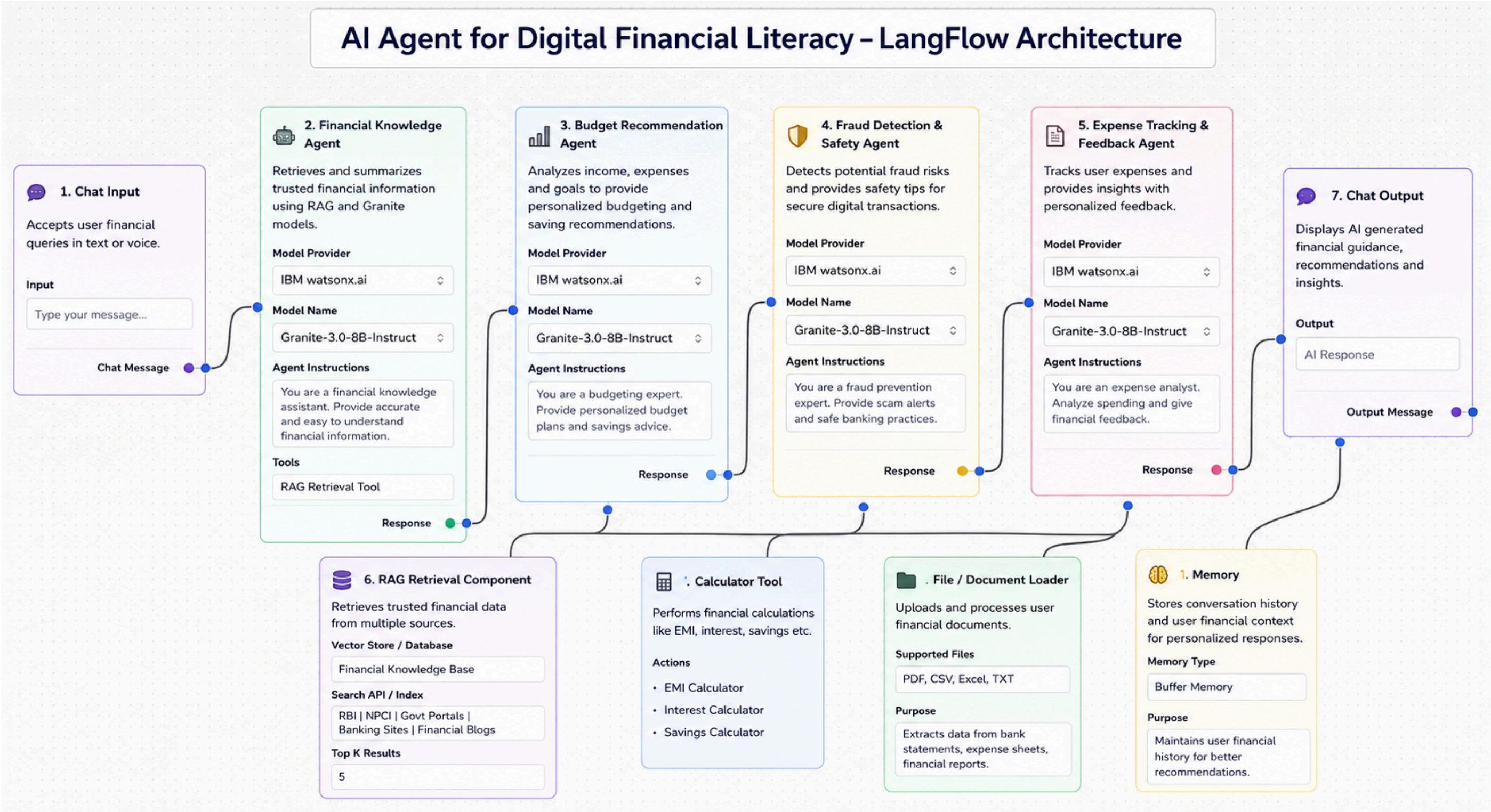
LangFlow is used to orchestrate agent workflows, while watsonx.ai ensures scalable AI deployment, embeddings, and AI governance. Granite models power reasoning, multilingual interaction, summarization, and personalized financial recommendations.

Overall, this solution delivers an end-to-end intelligent financial literacy assistant that promotes safe digital financial practices, improves financial awareness, protects users from fraud, and empowers individuals to make smarter financial decisions with confidence.

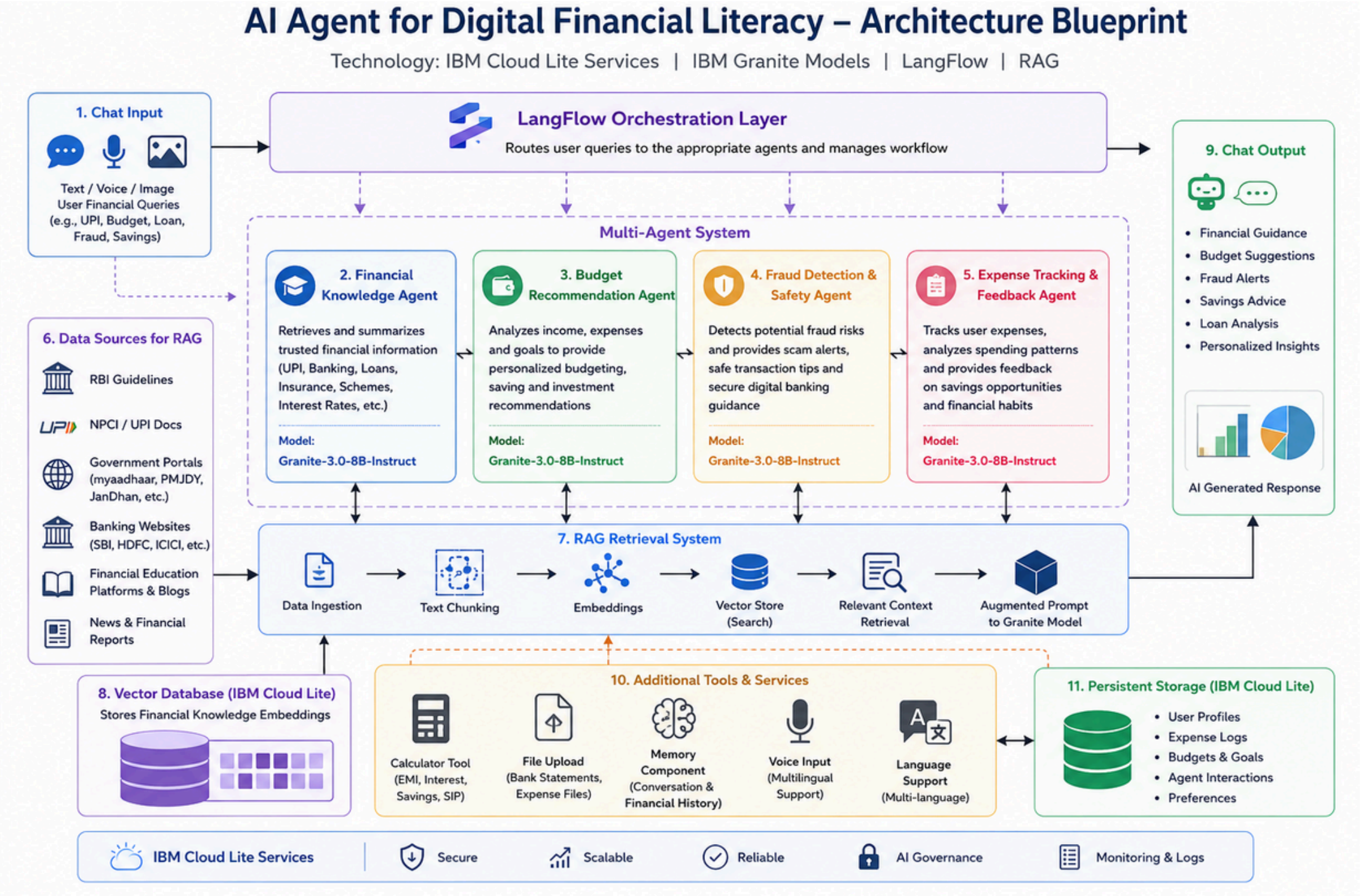
Langflow component Used -

1. Chat Input – Interface where users enter their financial queries, spending details, banking concerns, or budgeting information through text or voice interaction.
2. IBM Cloud Lite Services + IBM Granite Models – Processes user input using IBM Granite models to generate personalized financial guidance, fraud prevention tips, budgeting suggestions, and digital banking assistance.
3. Name of IBM Granite Model – Granite-3.0-8B-Instruct → Efficient for real-time financial literacy recommendations, multilingual support, and intelligent financial conversations.
4. Chat Output – Displays AI-generated responses such as UPI guidance, budgeting advice, loan interest analysis, fraud alerts, savings recommendations, and financial education insights.
5. File Component – Stores and retrieves user expense records, financial datasets, budgeting history, and banking-related information for analysis and personalized recommendations.
6. Retrieval-Augmented Generation (RAG) Component – Retrieves trusted financial information from banking websites, RBI guidelines, government portals, and educational resources to provide accurate and reliable responses.

Langflow workflow -



Architecture blueprint



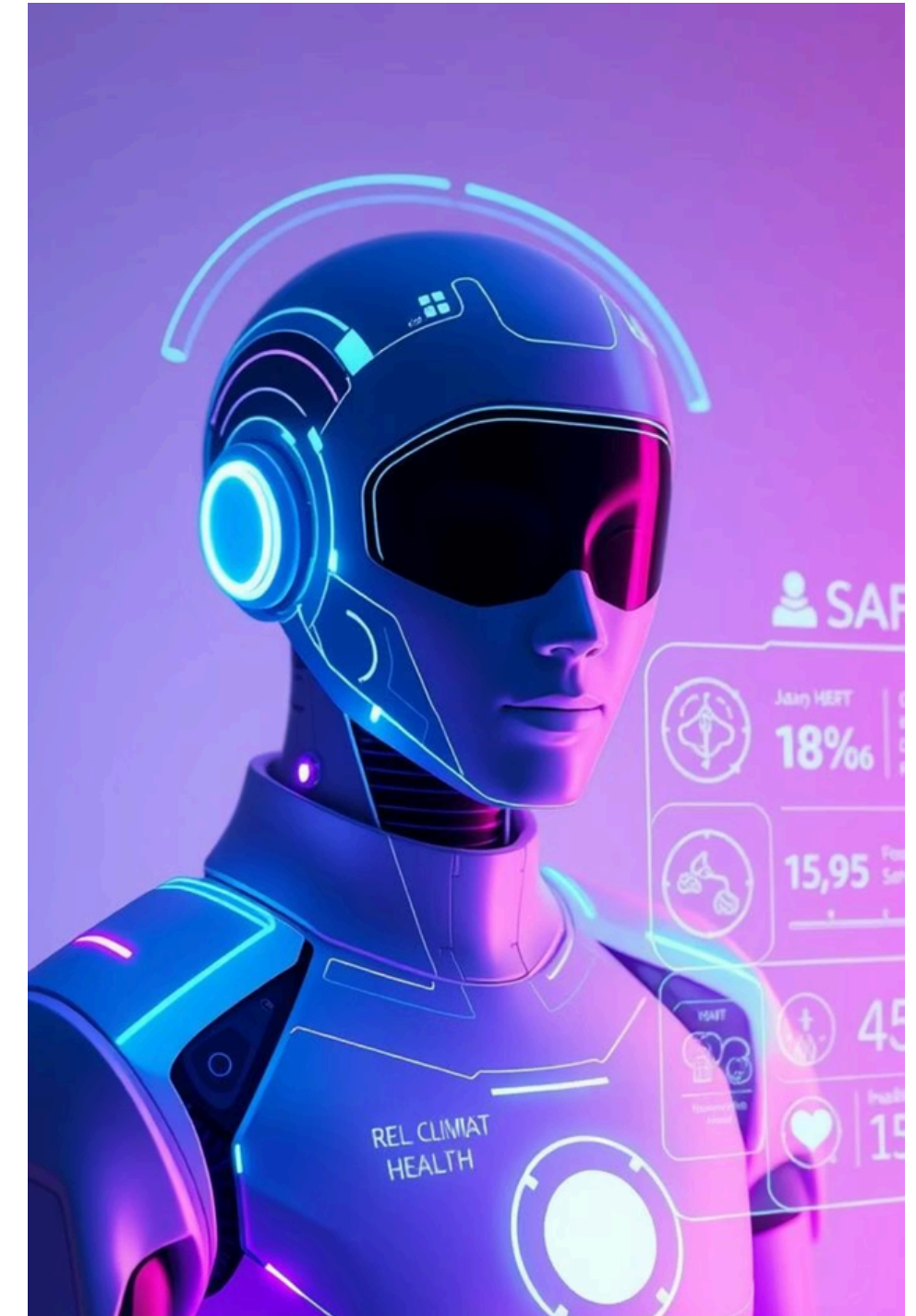
Role of Agentic AI in the solution

Role of Agentic AI in Digital Financial Literacy Agent

Agentic AI enables the Digital Financial Literacy Agent to function as an intelligent, autonomous financial assistant rather than a basic chatbot. Using IBM Cloud Lite Services and IBM Granite models, the system coordinates multiple specialized agents to deliver personalized financial guidance and digital banking support.

Each agent performs a specific role—such as retrieving trusted financial information, generating budgeting recommendations, providing fraud prevention guidance, and analyzing user expenses—while collaboratively working together. This multi-agent architecture ensures efficient task handling, accurate responses, and smarter financial decision-making. Agentic AI also provides context awareness by understanding user information such as income, spending habits, savings goals, and financial concerns to generate personalized recommendations. It supports real-time adaptation by continuously updating financial suggestions based on user interactions, expense tracking, and changing financial needs. Additionally, agents communicate with one another to improve response quality, risk detection, and budgeting insights, enabling more effective and intelligent financial assistance.

Overall, Agentic AI makes the system proactive, adaptive, personalized, and goal-driven, acting like a smart digital financial advisor that helps users manage money safely, avoid fraud, improve financial literacy, and make confident financial decisions.



Technology Used

- LangFlow Platform
- Used for visually designing and orchestrating multi-agent workflows for the Digital Financial Literacy Agent, enabling smooth coordination between financial knowledge, budgeting, fraud detection, and expense tracking agents.
- IBM Granite Model – Granite-3.0-8B-Instruct
 - Used for natural language understanding, reasoning, multilingual interaction, financial query handling, and personalized financial recommendations.
- IBM Cloud Lite Services
 - Provides the infrastructure to build, deploy, and scale the Digital Financial Literacy Agent efficiently. It ensures a secure, reliable, and scalable environment for AI deployment and data management.
- Retrieval-Augmented Generation (RAG)
 - Used to retrieve trusted financial information from RBI guidelines, banking websites, NPCI resources, and government financial portals before generating intelligent and accurate responses.
- IBM watsonx.ai
 - Used for AI model deployment, embeddings, orchestration, governance, and managing intelligent financial interactions powered by IBM Granite models.
- Vector Database (FAISS/Chroma)
 - Used for storing and retrieving financial knowledge embeddings to support fast and accurate RAG-based financial information retrieval.

Novelty and Uniqueness

- The proposed Digital Financial Literacy Agent stands out by combining Agentic AI, RAG, and personalized financial guidance into a single intelligent system.
- Multi-Agent Intelligence
- Instead of using a single chatbot, multiple specialized agents collaborate to provide accurate, context-aware financial assistance related to budgeting, digital payments, fraud prevention, savings, and banking literacy.
- RAG-Based Accuracy
- Integrates real-time financial data retrieval with AI-generated responses, reducing misinformation and ensuring trustworthy guidance from RBI guidelines, banking portals, and government financial resources.
- Hyper-Personalization
- Tailors financial recommendations based on user income, spending patterns, savings goals, financial behavior, and personal preferences.
- Real-Time Financial Feedback Loop
- Continuously adapts budgeting suggestions, savings advice, and fraud alerts based on user expenses, financial activities, and interaction history.
- Multi-Modal Input
- Supports text and voice-based interactions for easier accessibility and improved user experience across diverse user groups.
- Financial Safety & Awareness Focus
- Goes beyond financial education by providing scam detection guidance, secure digital payment practices, fraud prevention tips, and safe online banking recommendations.
- Scalable & Enterprise-Ready
- Built using IBM Cloud Lite Services and IBM Granite Models, ensuring scalability, reliability, AI governance, and secure deployment for large-scale financial literacy applications.
- This makes the solution a smart, adaptive, and proactive digital financial advisor, not just a static financial information chatbot.

Git Hub Link

· Please create public github repository (Enable Add README button while creating repository)

with format –

Please test and Paste the working Github URL –<https://github.com/Madhulahari2526/IBM>

Future Scope

1. Integration with Banking & Financial Platforms

The system can be integrated with digital banking services, UPI platforms, budgeting applications, and financial APIs to collect real-time financial data such as expenses, savings, transaction history, and payment activities. This will enable the Digital Financial Literacy Agent to provide more accurate budgeting suggestions, spending analysis, fraud alerts, and personalized financial recommendations based on actual user financial behavior.

2. AI-Powered Voice Assistant & Multilingual Support

Enhancing the system with a voice-enabled assistant and multilingual support will improve accessibility and financial inclusion. Users can interact naturally in their preferred language to ask questions about digital payments, banking, loans, budgeting, and fraud prevention. This makes the solution more user-friendly, inclusive, and suitable for people from diverse linguistic and educational backgrounds.

Thank You!

Thank you for your time and interest.